

MUHAMMAD MUSAWAR ALI SHAH

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PROFESSIONAL SUMMARY

Third-year Computer Science student at FAST-NUCES Karachi with hands-on experience in full-stack web development, AI/Machine Learning, and object-oriented software design using Python, C++, and Java. Built role-based web and desktop applications with PostgreSQL, FastAPI, React, and Java Swing, and completed coursework in machine learning models, search algorithms, and data structures. Microsoft Learn Student Ambassador (2025–26) and IEEE member experienced in explaining technical concepts to peers. Seeking an internship or junior developer role in software development or AI/ML.

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES), Karachi Campus	2024 – Present
Bachelor of Science in Computer Science (BS-CS), 3rd Year (transferred from Peshawar Campus)	
H.S.S.C (Intermediate / A-Level equivalent) – Federal Board (FBISE), Grade: A	2021 – 2023
S.S.C (Matriculation / O-Level equivalent) – Federal Board (FBISE), Grade: A-1	2019 – 2021

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, JavaScript (Basic)
AI / Machine Learning: Linear and Logistic Regression, Gradient Descent, Naive Bayes, K-Means, K-NN, Neural Networks, Search Algorithms (BFS, DFS, UCS, A*, Hill Climbing, Simulated Annealing), Adversarial Search (Minimax, Alpha-Beta Pruning), Constraint Satisfaction
ML Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, NetworkX; familiar with TensorFlow, PyTorch, OpenAI APIs
Web & Backend: React, FastAPI, REST APIs, Axios, HTML5, CSS3
Databases: PostgreSQL, MySQL, SQLite
GUI & Game Development: Qt, SFML, Java Swing
Tools & Platforms: Git, GitHub, Linux (Ubuntu, Arch), VS Code, IntelliJ IDEA, Eclipse, MS Office
Core Concepts: Object-Oriented Programming, Data Structures, Algorithms, SDLC, Software Design & Analysis, Role-Based Access Control

PROJECTS

Lost & Found Intelligence System React, FastAPI, PostgreSQL	2025
- Built a full-stack web app digitizing the university Lost & Found workflow (replacing a manual register), with item registration, search, claim review, digital receipts, and in-app notifications - Designed a 9-table PostgreSQL schema with custom ENUMs, cascade deletes, and audit logging - Implemented role-based access control (Admin, Staff, Student, Faculty) with SHA-256 password hashing and parameterized queries to prevent SQL injection	
IT Components Problem Reporting System Java, Swing, OOP	2025
- Developed a role-based desktop complaint management system for Students, Faculty, IT Staff, and Admins with complaint submission, priority assignment, status tracking, and reports - Applied layered architecture (model, repository, service, UI) and enum-driven state machines for complaint lifecycles	
Chess Game and Snake Game C++, OOP, Qt, SFML	2024
Built two GUI games using OOP: Chess with complete rules, piece movement, and turn-based play (Qt); Snake with real-time mechanics and collision detection (SFML)	
Weather Application JavaScript, REST API	2024
Built a web application that fetches and displays real-time weather data from a public REST API based on user input	

LEADERSHIP & ACTIVITIES

Microsoft Learn Student Ambassador (MLSA) – Microsoft	2025 – 2026
Selected for Microsoft's global student leadership program; organized developer workshops and peer-learning sessions and promoted Microsoft developer and AI tools on campus	
Student Member – IEEE	2024 – Present
Active participant in technical events, workshops, and student activities in computing and engineering	

CERTIFICATIONS

Video Editing, Animation & Vlogging – DigiSkills Training Program (DSTP 3.0, Batch-01)	Aug – Nov 2025
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RELEVANT COURSEWORK

Object-Oriented Programming, Data Structures, Algorithms, Database Systems, Artificial Intelligence, Operating Systems, Software Design & Analysis, Discrete Structures, Linear Algebra, Probability & Statistics, Theory of Automata, Computer Organization & Assembly Language